

Subnet cheat sheet

Subnet mask, wildcard, total addresses and usable hosts for /0 to /32.

Prefix	Subnet mask	Wildcard	Addresses	Usable hosts	/24s
/0	0.0.0.0	255.255.255.255	4,294,967,296	4,294,967,294	16,777,216
/1	128.0.0.0	127.255.255.255	2,147,483,648	2,147,483,646	8,388,608
/2	192.0.0.0	63.255.255.255	1,073,741,824	1,073,741,822	4,194,304
/3	224.0.0.0	31.255.255.255	536,870,912	536,870,910	2,097,152
/4	240.0.0.0	15.255.255.255	268,435,456	268,435,454	1,048,576
/5	248.0.0.0	7.255.255.255	134,217,728	134,217,726	524,288
/6	252.0.0.0	3.255.255.255	67,108,864	67,108,862	262,144
/7	254.0.0.0	1.255.255.255	33,554,432	33,554,430	131,072
/8	255.0.0.0	0.255.255.255	16,777,216	16,777,214	65,536
/9	255.128.0.0	0.127.255.255	8,388,608	8,388,606	32,768
/10	255.192.0.0	0.63.255.255	4,194,304	4,194,302	16,384
/11	255.224.0.0	0.31.255.255	2,097,152	2,097,150	8,192
/12	255.240.0.0	0.15.255.255	1,048,576	1,048,574	4,096
/13	255.248.0.0	0.7.255.255	524,288	524,286	2,048
/14	255.252.0.0	0.3.255.255	262,144	262,142	1,024
/15	255.254.0.0	0.1.255.255	131,072	131,070	512
/16	255.255.0.0	0.0.255.255	65,536	65,534	256
/17	255.255.128.0	0.0.127.255	32,768	32,766	128
/18	255.255.192.0	0.0.63.255	16,384	16,382	64
/19	255.255.224.0	0.0.31.255	8,192	8,190	32
/20	255.255.240.0	0.0.15.255	4,096	4,094	16
/21	255.255.248.0	0.0.7.255	2,048	2,046	8
/22	255.255.252.0	0.0.3.255	1,024	1,022	4
/23	255.255.254.0	0.0.1.255	512	510	2
/24	255.255.255.0	0.0.0.255	256	254	1
/25	255.255.255.128	0.0.0.127	128	126	1/2
/26	255.255.255.192	0.0.0.63	64	62	1/4
/27	255.255.255.224	0.0.0.31	32	30	1/8
/28	255.255.255.240	0.0.0.15	16	14	1/16
/29	255.255.255.248	0.0.0.7	8	6	1/32
/30	255.255.255.252	0.0.0.3	4	2	1/64
/31	255.255.255.254	0.0.0.1	2	2	1/128
/32	255.255.255.255	0.0.0.0	1	1	1/256

Magic number, /25 to /30

Prefix	Last octet	Block size	Subnet starts
/25	128	128	0, 128
/26	192	64	0, 64, 128, 192
/27	224	32	0, 32, 64, 96, 128, 160, 192, 224
/28	240	16	0, 16, 32, 48, ... 240
/29	248	8	0, 8, 16, 24, ... 248
/30	252	4	0, 4, 8, 12, ... 252

Hosts needed to prefix

Hosts needed	Prefix	Subnet mask	Usable hosts
2	/30	255.255.255.252	2
6	/29	255.255.255.248	6
14	/28	255.255.255.240	14
30	/27	255.255.255.224	30
62	/26	255.255.255.192	62
126	/25	255.255.255.128	126
254	/24	255.255.255.0	254
510	/23	255.255.254.0	510